Liu Jason Tan

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EDUCATION

Columbia University — New York, NY
Master of Science in Computer Science — Expected 2030
Track: Vision, Graphics, Interaction, and Robotics

University of Michigan - Ann Arbor — Ann Arbor, MI
Master of Applied Data Science, GPA: 4.00 /4.00 — Aug 2022
Relevant Courses: Data Mining, Supervised Learning, Unsupervised Learning, Cloud Computing

Stony Brook University — Stony Brook, NY
Bachelor of Science in Information Systems, GPA: 3.64 /4.00 — May 2020
Relevant Courses: Object-Oriented Programming, Data Structures, Database Design

WORK EXPERIENCE

Morgan Stanley — New York, NY
Associate, Operational Risk Capital Analytics (Full Time) — Aug 2022 - Present

- Reconstructed a critical operational risk capital model from scratch, leveraging Python to replace a legacy implementation with 10k+ lines of clean, efficient code.
- Optimized natural language processing and machine learning models to enhance operational risk incident quality assurance, reducing manual review workload by over 50% and increasing data accuracy for risk event analysis.
- Served as a subject matter expert in capital reporting by collaborating with global teams to achieve strategic goals and deliver robust results that exceeded stakeholders' and regulators' expectations.
- Led and coached 2 team members and 5 consultants by setting clear guidelines, managing timelines, and running daily standups, enabling the team to deliver mission-critical capital analytics solutions ahead of regulatory deadlines.

Poisera — Remote
Data Analytics Intern — Jun 2021 - Aug 2021

- Applied web scraping and API integration pipelines to collect and analyze large-scale public datasets, directly informing product development decisions, leading to significant user interface improvements.
- Conducted user interviews and qualitative research to identify critical insights, driving key management decisions and boosting customer satisfaction.

ACADEMIC PROJECTS

- **MyVoice Data Challenge (First Place)** - Engineered an NLP pipeline to analyze sentiment in text message surveys regarding COVID-19. Automated data cleaning, text encoding, and hierarchical clustering employing BERT to improve the efficiency of research to generate deeper insights.
- **S&P 500 Stock Performance Forecasting** - Achieved 62% precision with a random forest classifier, a substantial improvement over the 20% precision of a dummy classifier. Categorized stocks into top, middle, and bottom tiers using key equity metrics such as price-to-earnings ratio, dividend yield, and volatility.
- **Social Media Monitoring** - Developed a full-stack application for real-time sentiment and topic monitoring of company discussions. Utilized supervised and unsupervised learning techniques, including BERT for emotion classification (e.g., surprise, anger, disgust) and non-negative matrix factorization for topic clustering (e.g., account issues, ordering issues, service issues), to gain actionable insights from social media interactions.

SKILLS

Programming languages: Python, Java, MATLAB, R, SQL
Python Packages: NumPy, Pandas, PyTorch, TensorFlow, SciKit Learn, SciPy, Matplotlib, PySpark, OpenCV
Tools: Jupyter Notebook, GitHub, Bitbucket, Jira, API, Agile Workflows, Co-Pilot
Machine Learning: Regression, deep neural networks, classification, clustering, Large Language Models